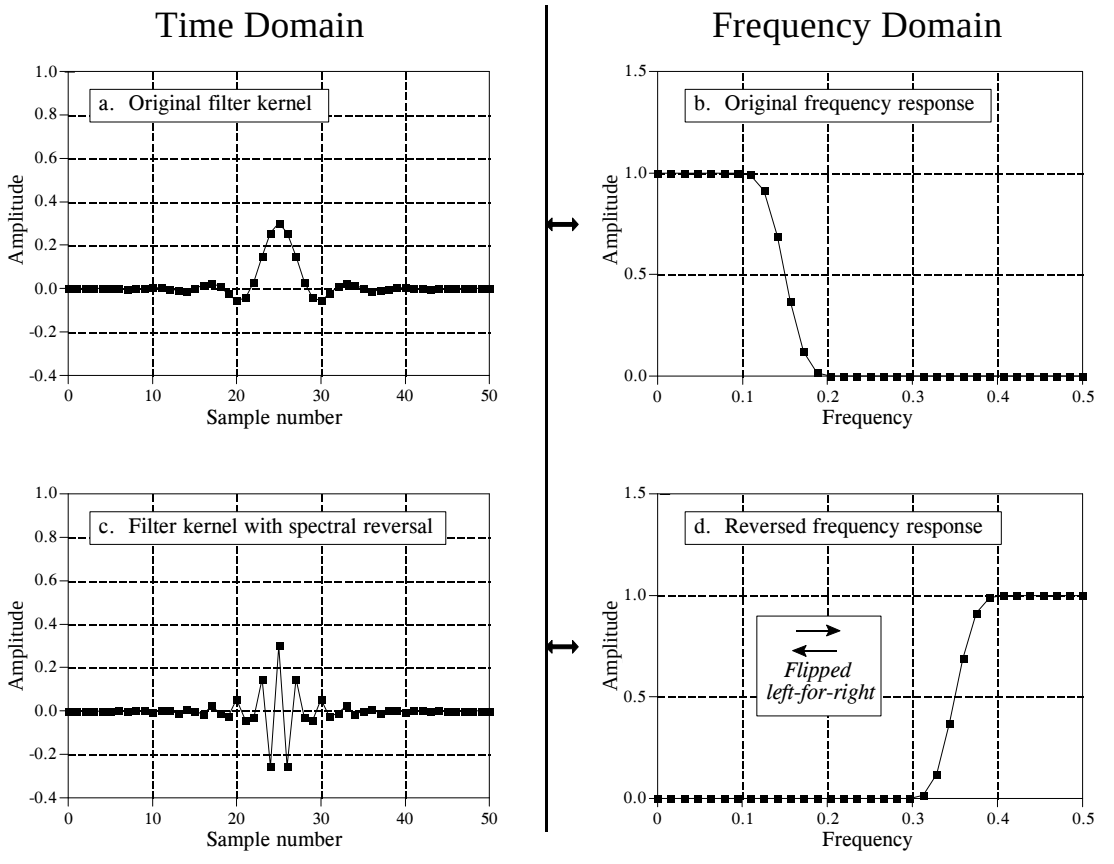

FIGURE 14-7
Example of spectral reversal. The low-pass filter kernel in (a) has the frequency response shown in (b). A high-pass filter kernel, (c), is formed by changing the sign of every other sample in (a). This action in the time domain results in the frequency domain being flipped *left-for-right*, resulting in the high-pass frequency response shown in (d).

7, parallel systems with added outputs can be combined into a single stage by adding their impulse responses. As shown in (b), the filter kernel for the high-pass filter is given by: $\delta[n] - h[n]$. That is, change the sign of all the samples, and then add one to the sample at the center of symmetry.

For this technique to work, the low-frequency components exiting the low-pass filter must have the same phase as the low-frequency components exiting the all-pass system. Otherwise a complete subtraction cannot take place. This places two restrictions on the method: (1) the original filter kernel must have left-right symmetry (i.e., a zero or linear phase), and (2) the impulse must be added at the center of symmetry.

The second method for low-pass to high-pass conversion, *spectral reversal*, is illustrated in Fig. 14-7. Just as before, the low-pass filter kernel in (a) corresponds to the frequency response in (b). The high-pass filter kernel, (c), is formed by *changing the sign of every other sample* in (a). As shown in (d), this flips the frequency domain *left-for-right*: 0 becomes 0.5 and 0.5